

Contents

1	Introduction: Fermat's Last Theorem	2
2	Truncated Euler Systems	4
3	Properties of Truncated Euler Systems	6
4	The Main Result	11
5	Next Steps	13
A	Algebraic Number Theory	14

List of Symbols

$\mathbf{Z}_p$	p -adic integers	3
C_K	Cyclotomic units	3
E_K	Global units	3
ζ_n	Primitive n^{th} root of unity	4
μ_n	n^{th} roots of unity	4
G_K	$\text{Gal}(\overline{K}/K)$	4
G_r	$\text{Gal}(K(\mu_n)/K)$	4
I_K^n	See page 4	4
$I_{K,M}^n$	$I_K^n \cap I_{\mathbf{Q}(\mu_M)}^n$	4
I^n	Same as I_K^n	4
I_M^n	Same as $I_{K,M}^n$	4
$\mathcal{E}_K^n$	truncated Euler system of depth n	4
S_K^n	Special units of depth n	4
$\mathcal{E}_M^n$	truncated Euler systems defined on I_M^n	4
$\mathbf{N}_r$	norm operator of G_r	4
Fr_l	Frobenius of l	4
$\mathbf{D}_l$	Kolyvagin derivative	4
$e(\chi)$	χ -idempotent	11
$A(\chi)$	χ -eigenspace of A	11
ε	Generator of $(E_K/C_K)(\chi)$	11
T_l	Inertia field of l	12
I_l	Inertia group of l	12
Z_l	Decomposition field of l	12
G_l	Decomposition group of l	12
Cl_K	Ideal class group of K	14

Abstract

The interaction of the units of a number field and its ideal class group has long been a topic of study in algebraic number theory. In the 1980s, Rubin discovered the notion of “special units” and it was conjectured in [Seo08] that they encode information about the exponent of the class group. Expanding on these ideas, Kolyvagin introduced the machinery of Euler systems, and they were subsequently used by Rubin to provide a new proof of the Main Conjecture of Iwasawa theory. We aim to make clearer and expand some of the arguments given in [Rub90], culminating in a reformulated version of lemma 2.6 in [Seo08].

1 Introduction: Fermat's Last Theorem

The motivation for algebraic number theory begins with the study of *Diophantine equations*. The Pythagorean theorem, known from Babylonian times, states that the lengths of the sides of a right triangle in Euclidean space satisfy $a^2 + b^2 = c^2$. One may wonder whether there are right triangles with integer side lengths; algebraically, we ask which *integers* satisfy the Pythagorean equation. Of course, any tuple $(a, b, c) = (0, x, x)$ will trivially satisfy this; we thus ask whether there are *nontrivial* integral solutions to the equation $a^2 + b^2 = c^2$. There are in fact, infinitely many *distinct* integer solutions to this Diophantine equation (in fact, in the language of arithmetic geometry, the group of rational points on the circle is not even finitely generated).

Similarly to the Pythagorean equation, we can study integral solutions to the equation $a^n + b^n = c^n$ for a fixed positive integer n . As before, setting one of a, b to be 0 admits a large family of solutions; we thus ask again for nontrivial solutions to the equation. Pierre de Fermat conjectured in 1637 what would become known as “Fermat's Last Theorem”: that there were in fact no nontrivial solutions to this equation for any $n > 2$ ¹. Early on, the following reductions were realized:

1. Since any solution to $a^{km} + b^{km} = c^{km}$ is a solution to $a^k + b^k = c^k$, it suffices to prove the theorem for prime exponents and for $n = 4$.
2. If $(ar)^n + (br)^n = (cr)^n$ for some $r \in \mathbf{Z}$, then $a^n + b^n = c^n$; we can thus assume that a, b, c are coprime.

Thus for each odd prime and $n = 4$, there are two cases to prove: the first case, in which $n \nmid abc$, and the second case in which $n \mid abc$. The first case is easier to treat than the second, and we describe the ideas and efforts that historically went into proving the first case here. For low exponents, the theorem (in both cases) was proven by Fermat, Euler, Gauss, and others. Perhaps the first major attempt to prove Fermat's Last Theorem came from Sophie Germain in the early nineteenth century; her eponymous theorem allowed for the proof of the first case for all sufficiently small primes.

More directly relevant to the history of algebraic number theory was Gabriel Lamé's idea to factor the equation $x^p + y^p = z^p$ in the ring $\mathbf{Z}[\zeta_p]$. Assuming the existence of a first case nontrivial solution to the Fermat equation along with $\mathbf{Z}[\zeta_p]$ being a unique factorization domain, one may arrive at a contradiction which implies Fermat's Last Theorem holds in the first case for all prime exponents. The crucial flaw in this idea was the assumption that $\mathbf{Z}[\zeta_p]$ is a unique factorization domain. Indeed, the only p for which this holds are $p = 3, 5, 7, 11, 13, 17, 19$ so Lamé's idea, it seems, falls far short of what was hoped. Fortunately, the technique could be recovered to prove the theorem for a larger class of primes. Ernst Kummer's idea was to use what he called “ideal numbers”² in order to recover unique factorization for the ring $\mathbf{Z}[\zeta_p]$. His work culminated in a proof³ of the first case of Fermat's Last Theorem for “regular primes”⁴.

While Kummer's work in some regards falls far short of proving Fermat's Last Theorem⁵, it sets the stage for the

¹When Fermat wrote down the conjecture, he wrote “I have discovered a truly marvelous proof of this, which this margin is too narrow to contain.” While a proof of the $n = 4$ case is attributed to him, the subsequent 358 years of mathematical efforts which included the development of the fields of algebraic number theory and algebraic geometry makes it unlikely that Fermat had a valid proof of the full theorem.

²In modern terms, ideal numbers are generators of principal ideals.

³A full treatment of this proof, as well as a proof of the second case for certain primes can be found in [Was82].

⁴It is conjectured that there are infinitely many regular primes, of relative density $e^{-1/2}$, but this is still open. It is known only that there are infinitely many irregular primes.

⁵Fermat's Last Theorem was finally resolved in 1995 with Andrew Wiles' proof of the Modularity Theorem for semistable elliptic curves [Wil95],[TW95]. His work with Richard Taylor built on the efforts of Serre, Frey, and Ribet [Rib90],[Fre86] and ultimately led to the proof of the full Modularity Theorem by Breuil, Conrad, Diamond, and Taylor [BCDT01].

introduction and development of algebraic number theory as its own area of study. Kummer's use of ideal numbers and the condition of regularity motivates the study of the ideal class group, the further history of which we summarise in brief. In 1959, Iwasawa began the study of $\mathbf{Z}_p$ -extensions of number fields [Iwa59] and provided an expression for the exponent of the p -primary part of Galois modules in these extensions. Ferrero and Washington proved [Was82] that Iwasawa's μ -invariant vanishes for the ideal class groups in cyclotomic $\mathbf{Z}_p$ -towers of abelian number fields; an alternative proof of this was later given by Sinnott. For any abelian number field, one may define (by virtue of the Kronecker-Weber theorem) the "cyclotomic units" C_K . Sinnott proved that the index of these units in the full unit group E_K (also written as $\mathcal{O}_K^\times$) has a direct relation to the class number of K [Sin80]. In the same vein, Thaine [Tha88] and later Rubin [Rub87] introduced and studied what are now called "special units". This report is dedicated to the study of these special units, particularly through the machinery of Euler systems⁶.

We begin with an introduction to the basic concepts of truncated Euler Systems, as defined in [Seo08]; we then provide proofs of certain properties of truncated Euler systems (RI, RII, RIV in [Seo08]). These are proven in [Rub90] in a special case, and we adapt these proofs to the generalized setting established by Seo. The culmination of this report is **lemma 4** and **theorem 3**, which together are a simplification and strengthening of lemma 2.6 in [Seo08].

Acknowledgements

I would like to thank Dr. Timothy All for his supervision of this thesis and his guidance through the past three years. I would also like to thank Connor Lane for our insightful conversations on this topic, which greatly contributed to the final form of this report.

⁶First introduced by Victor Kolyvagin [Kol90], Euler systems have been used by him and later Karl Rubin to make major progress on the celebrated Birch and Swinnerton-Dyer Conjecture. Rubin also used Euler systems in his proof of the Main Conjecture of Iwasawa theory [Rub90], simplifying the arguments originally given by Mazur and Wiles [MW84]. Euler systems were also invaluable to Wiles' efforts on the Modularity Theorem, and played a major role in his resolution to Fermat's Last Theorem.

2 Truncated Euler Systems

We assume that the reader is familiar with the basic notions of algebraic number theory and Galois cohomology. Algebraic number theory we summarise in the appendix, while we refer the reader to [NSW13] or [Ked] for Galois cohomology. We establish the following notation consistent with [Seo08]:

Let K be a real abelian number field and let p be an odd prime. We let ζ_n denote a primitive n^{th} root of unity and $\mu_n = \langle \zeta_n \rangle$ the group of n^{th} roots of unity. Additionally, we let $G_K := \text{Gal}(\bar{K}/K)$ and for any positive integer r let $G_r := \text{Gal}(K(\mu_r)/K)$. We define

$$I_K^n := \{p_1 \cdots p_m : m \leq n, p_i > 0 \text{ prime}, p_i \neq p_j (i \neq j), p_i \text{ splits completely in } K\}.$$

We note that $1 \in I_K^n$ as the empty product of primes. For M a positive odd integer, we let

$$I_{K,M}^n := \{a \in I_K^n : \forall l \mid a (l \text{ prime}), l \equiv 1 \pmod{M}\} = I_K^n \cap I_{\mathbf{Q}(\mu_M)}^n.$$

For convenience, we will frequently omit the K dependence on these sets, writing I^n and I_M^n respectively.

We define the “truncated Euler systems” $\mathcal{E}_K^n$ as the set of maps $\psi : I^n \rightarrow \bar{\mathbf{Q}}$ satisfying the following conditions for $m, r \in I^n$ with $m \mid r$:

1. $\psi(m) \in K(\zeta_m^+)$ where we write $K(\zeta_m^+)$ for $K(\zeta_m + \zeta_m^{-1})$;
2. $\text{Norm}_{K(\zeta_r^+)/K(\zeta_m^+)} \psi(r) = \psi(m)^{\prod_{p \mid r, p \nmid m} (1 - \text{Fr}_p^{-1})}$ where Fr_p denotes the Frobenius of p in the extension $K(\zeta_m^+)/\mathbf{Q}$;
3. If $p \nmid m$ then $\psi(mp) \equiv \psi(m) \pmod{\mathfrak{P}}$ for all primes $\mathfrak{P}$ of $K(\zeta_{mp}^+)$ over p .

With these we define the special units of depth n , $S_K^n := \{\psi(1) : \psi \in \mathcal{E}_K^n\} \cap E_K$. We note that the depth 1 special units, S^1 , coincide with the special units defined in [Rub87]. As with I_K^n we will omit the K subscript and write $\mathcal{E}^n$ and S^n for convenience. We similarly define $\mathcal{E}_M^n$ as the maps $\psi : I_M^n \rightarrow \bar{\mathbf{Q}}$ satisfying the same conditions as those in $\mathcal{E}^n$, and S_M^n as the image of 1 in E_K .

For $r \in I^n$, let $\mathbf{N}_r$ denote the norm operator given by

$$\mathbf{N}_r := \sum_{\sigma \in G_r} \sigma \in \mathbf{Z}[G_r].$$

As r is squarefree, we have a canonical isomorphism of Galois groups $G_r = \prod_{l \mid r} G_l$ and thus

$$\mathbf{N}_r = \prod_{l \mid r} \mathbf{N}_l.$$

If $l \in I^n$ and $l \nmid r$, G_l is canonically isomorphic to $\text{Gal}(F(\mu_r)/F(\mu_r))$ and we will often identify G_l with this Galois group. We write Fr_l for the Frobenius of l in G_r .

For each $l \in I^n$, fix a generator σ_l of $G_l = (\mathbf{Z}/l\mathbf{Z})^\times$ and define

$$\mathbf{D}_l := \sum_{i=1}^{l-2} i \sigma_l^i \in \mathbf{Z}[G_l].$$

Proposition 1

The operator $\mathbf{D}_l$ satisfies the identity $(\sigma_l - 1)\mathbf{D}_l = (l - 1) - \mathbf{N}_l$.

Proof. Observe that

$$(\sigma_l - 1)\mathbf{D}_l = (\sigma_l - 1) \sum_{i=1}^{l-2} i \sigma_l^i = \sum_{i=2}^{l-1} (i - 1) \sigma_l^i - \sum_{i=1}^{l-2} i \sigma_l^i = (l - 1) \sigma_l^{l-1} - \sum_{i=1}^{l-1} \sigma_l^i = (l - 1) - \mathbf{N}_l.$$

More generally, for $r \in I^n$, we define

$$\mathbf{D}_r := \prod_{l|r} \mathbf{D}_l.$$

We give a concrete example of an Euler system, in the case that $F = \mathbf{Q}(\mu_m)^+$. Let ζ_m be a primitive m^{th} root of unity and ζ_l a primitive l^{th} root of unity (l a prime in I^n). For $r \in I^n$ define

$$\xi_r := \left(\zeta_m \prod_{l|r} \zeta_l - 1 \right) \left(\zeta_m^{-1} \prod_{l|r} \zeta_l - 1 \right).$$

Theorem 1

Let $r \in I_{F,M}^n$ with $r > 1$ and $l \in I_{F,M}^1$ such that $l \mid r$ (and $l > 1$). The algebraic integer ξ_r satisfies the following conditions:

ES 1. $\xi_r \in F(\mu_r)^\times$.

ES 2. ξ_r is a cyclotomic unit.

ES 3. $\xi_r^{\mathbf{N}_l} = \xi_{r/l}^{(\text{Fr}_l - 1)}$.

ES 4. $\xi_r \equiv \xi_{r/l} \pmod{\mathfrak{l}}$ for every prime $\mathfrak{l}$ above l .

In particular, the map $r \mapsto \xi_r$ defines an Euler system in $\mathcal{E}_M^n$.

Proof.

ES 1. Recall that $F \subset \mathbf{Q}(\mu_m)$ and $F(\mu_r) = F(\mu_{l_1}) \cdots F(\mu_{l_j})$ where $r = l_1 \cdots l_j$. From this, it follows that $\xi_r \in \mathbf{Q}(\mu_{rm})$. Observe then that the action $\zeta_m \mapsto \zeta_m^{-1}$ fixes ξ_r , so $\xi_r \in F(\mu_r)^\times$.

ES 2. Note that $\zeta_m^{\pm 1} \prod_{l|r} \zeta_l$ is a primitive rm^{th} root of unity. Thus ξ_r is of the form $(\zeta_{rm} - 1)(\zeta_{rm}^b - 1)$ for some b coprime to rm , so ξ_r is a cyclotomic unit.

ES 3. We first need to determine (the actions of) the operators $\mathbf{N}_l \in \mathbf{Z}[\text{Gal}(F(\mu_r)/F(\mu_{r/l}))]$ and $\text{Fr}_l - 1 \in \mathbf{Z}[G_{r/l}]$. In particular, note that since $G_{r/l} = \text{Gal}(\mathbf{Q}(\mu_{r/l})/\mathbf{Q})$, Fr_l acts as the l^{th} power map on $\zeta_{r/l}$. We then have that

$$\begin{aligned} \xi_r^{\mathbf{N}_l} &= \prod_{k=1}^{l-2} \left(\zeta_l^k \zeta_m \prod_{p|r/l} \zeta_p - 1 \right) \left(\zeta_l^k \zeta_m^{-1} \prod_{p|r/l} \zeta_p - 1 \right) \\ &= \prod_{k=1}^{l-2} \left(\zeta_l^k \zeta_m \prod_{p|r/l} \zeta_p - 1 \right) \prod_{k=1}^{l-2} \left(\zeta_l^k \zeta_m^{-1} \prod_{p|r/l} \zeta_p - 1 \right). \end{aligned}$$

Each product is of the form $\prod_{k=1}^{l-2} (x \zeta_l^k - 1) = (x^l - 1)(x - 1)^{-1}$ so we have:

$$\begin{aligned} \xi_r^{\mathbf{N}_l} &= \left(\zeta_m^l \prod_{p|r/l} \zeta_p^l - 1 \right) \left(\zeta_m \prod_{p|r/l} \zeta_p - 1 \right)^{-1} \left(\zeta_m^{-l} \prod_{p|r/l} \zeta_p^l - 1 \right) \left(\zeta_m^{-1} \prod_{p|r/l} \zeta_p - 1 \right)^{-1} \\ &\quad \text{We have } l \equiv \pm 1 \pmod{m} \text{ so } \zeta_m^l = \zeta_m^{\pm 1} : \\ &= \left(\zeta_m \prod_{p|r/l} \zeta_p^l - 1 \right) \left(\zeta_m \prod_{p|r/l} \zeta_p - 1 \right)^{-1} \left(\zeta_m^{-1} \prod_{p|r/l} \zeta_p^l - 1 \right) \left(\zeta_m^{-1} \prod_{p|r/l} \zeta_p - 1 \right)^{-1} \\ &= \xi_{r/l}^{(\text{Fr}_l - 1)} \end{aligned}$$

ES 4. Recall that in $\mathbf{Q}(\zeta_l)$, the prime l factors as $(l) = (\zeta_l - 1)^{l-1}$. Thus, $\zeta_l \equiv 1 \pmod{\mathfrak{l}}$ for all primes $\mathfrak{l}$ above l . From this, it follows that $\xi_r \equiv \xi_{r/l} \pmod{\mathfrak{l}}$. ■

3 Properties of Truncated Euler Systems

Lemma 1

If $\psi \in \mathcal{E}_M^n$, then

$$\psi(r)^{\mathbf{D}_r} \in [K(\mu_r)^\times / (K(\mu_r)^\times)^M]^{G_r}$$

for all $r \in I_M^n$.

Proof. We proceed by induction on n . If $n = 0$, then $r = 1$ and $G_r = 1$ so the base case is satisfied. Suppose now that the lemma holds for some n , and let $r \in I_M^n$. Let l be a prime dividing r and note that $r/l \in I_M^{n-1}$. **Proposition 1** and the norm condition on Euler systems show that

$$\psi(r)^{(\sigma_l-1)\mathbf{D}_r} = \psi(r)^{((l-1)-\mathbf{N}_l)\mathbf{D}_{r/l}} = \psi(r/l)^{((l-1)-(\text{Fr}_l-1))\mathbf{D}_{r/l}} \equiv \psi(r/l)^{(1-\text{Fr}_l)\mathbf{D}_{r/l}} \pmod{(K(\mu_r)^\times)^{l-1}}.$$

Since $l \equiv 1 \pmod{M}$, we have that

$$\psi(r)^{(\sigma_l-1)\mathbf{D}_r} \equiv \psi(r/l)^{(1-\text{Fr}_l)\mathbf{D}_{r/l}} \pmod{(K(\mu_r)^\times)^M},$$

moreover, $\text{Fr}_l \in G_{r/l}$ so the action of Fr_l on $\psi(r/l)^{\mathbf{D}_{r/l}}$ is trivial by the induction hypothesis. Thus, it follows that

$$\psi(r/l)^{(1-\text{Fr}_l)\mathbf{D}_{r/l}} \in (K(\mu_{r/l})^\times)^M.$$

Since each $\sigma_l \in \text{Gal}(K(\mu_l)/K)$ generates G_r , the result follows. ■

Lemma 2

For every $r \in I_M^n$, we have

$$[K(\mu_r)^\times / (K(\mu_r)^\times)^M]^{G_r} \cong K^\times / (K^\times)^M.$$

From this lemma, we immediately derive the following:

Corollary 1

For every $\psi \in \mathcal{E}_M^n$, there exists a map $\kappa_\psi : I_M^n \rightarrow K^\times / (K^\times)^M$ such that

$$\kappa_\psi(r) \equiv \psi(r)^{\mathbf{D}_r} \pmod{(K(\mu_r)^\times)^M}$$

for all $r \in I_M^n$.

Proof of lemma 2. We will prove the following sequence of isomorphisms:

$$K^\times / (K^\times)^M \cong H^1(G_K, \mu_M) \cong H^1(G_{K(\mu_r)}, \mu_M)^{G_r} \cong [K(\mu_r)^\times / (K(\mu_r)^\times)^M]^{G_r}.$$

To that end, consider the Kummer exact sequence:

$$1 \longrightarrow \mu_M \longrightarrow \bar{K}^\times \xrightarrow{\cdot^M} \bar{K}^\times \longrightarrow 1$$

where the map $\cdot^M : \bar{K}^\times \rightarrow \bar{K}^\times$ is exponentiation by M . We induce a long exact sequence on cohomology

$$0 \longrightarrow \mu_M^{G_K} \longrightarrow K^\times \xrightarrow{\cdot^M} K^\times \longrightarrow H^1(G_K, \mu_M) \longrightarrow H^1(G_K, \bar{K}^\times) \xrightarrow{\cdot^M} H^1(G_K, \bar{K}^\times)$$

and note that $H^1(G_K, \bar{K}^\times) = 0$ by "theorem 90". Thus we have that

$$H^1(G_K, \mu_M) \cong K^\times / (K^\times)^M.$$

For the next isomorphism, we consider the inflation-restriction exact sequence:

$$0 \longrightarrow H^1\left(G_r, \mu_M^{G_K(\mu_r)}\right) \xrightarrow{\text{inf}} H^1(G_K, \mu_M) \xrightarrow{\text{res}} H^1(G_{K(\mu_r)}, \mu_M)^{G_r} \xrightarrow{\text{trans}} H^2\left(G_r, \mu_M^{G_K(\mu_r)}\right) \xrightarrow{\text{inf}} H^2(G_K, \mu_M)$$

As r is divisible only by primes congruent to 1 mod M , it follows that

$$\mu_M^{G_K(\mu_r)} = \mu_M \cap K(\mu_r) = \mu_M \cap K.$$

However, since K is real and M is odd, we have in fact that $\mu_M \cap K = 1$ and thus $H^i\left(G_r, \mu_M^{G_K(\mu_r)}\right) = 0$ for all i . It follows that restriction gives the isomorphism.

For the final isomorphism, we repeat the Kummer argument to get that

$$H^1(G_{K(\mu_r)}, \mu_M) \cong K(\mu_r)^\times / (K(\mu_r)^\times)^M.$$

This isomorphism is functorial and $G_{K(\mu_r)}$ -equivariant, so the final isomorphism follows as desired. $\blacksquare$

Let $\mathcal{O}_K$ denote the ring of integers of K and let $\mathcal{I} = \bigoplus_{\lambda} \mathbb{Z}\lambda$ be the group of fractional ideals of F (written additively). For each rational prime l , let $\mathcal{I}_l = \bigoplus_{\lambda|l} \mathbb{Z}\lambda$ so $\mathcal{I} = \bigoplus_l \mathcal{I}_l$. If $y \in K^\times$, let $(y) \in \mathcal{I}$ denote the principal ideal generated by y and $(y)_l \in \mathcal{I}_l$, $[y] \in \mathcal{I}/M\mathcal{I}$, $[y]_l \in \mathcal{I}_l/M\mathcal{I}_l$ as the projections of (y) . Note that $[y]$ and $[y]_l$ are well-defined modulo $(K^\times)^M$ as well.

Lemma 3

Let $G = \text{Gal}(K/\mathbb{Q})$; suppose l splits completely in K and $l \equiv 1 \pmod{M}$. There is a unique G -equivariant surjection

$$\varphi_l: (\mathcal{O}_K/l\mathcal{O}_K)^\times \twoheadrightarrow \mathcal{I}_l/M\mathcal{I}_l$$

which makes the following diagram commute:

$$\begin{array}{ccc} & K(\mu_l)^\times & \\ x \mapsto x^{1-\sigma_l} \swarrow & & \searrow x \mapsto [x^{1-N_l}]_l \\ (\mathcal{O}_K/l\mathcal{O}_K)^\times & \xrightarrow{\varphi_l} & \mathcal{I}_l/M\mathcal{I}_l \end{array}$$

where we identify $\mathcal{O}_{K(\mu_l)}/\lambda'$ with $\mathcal{O}_K/\lambda$ for each prime λ of K above l and λ' of $K(\mu_l)$ above λ .

Proof. The map $x \mapsto x^{1-N_l}$ surjects onto $\mathcal{I}_l$ since for any $\mathfrak{l}$, we can find $\mathfrak{m}$ coprime to $\mathfrak{l}$ such that $\mathfrak{l}\mathfrak{m}$ is a principal ideal. From this it follows that the right map in the diagram surjects onto $\mathcal{I}_l/M\mathcal{I}_l$. To show surjectivity of the left map, we first note that

$$(\mathcal{O}_K/l\mathcal{O}_K)^\times = \prod_{\mathfrak{l}|l} (\mathcal{O}_K/\mathfrak{l})^\times = \prod_{\mathfrak{l}|l} (\mathcal{O}_{\mathfrak{l}}/\mathfrak{l})^\times = \prod_{\mathfrak{l}|l} (\mathbb{Z}_l/\mathfrak{l})^\times$$

since l splits completely in K . Working locally, we have the map $\mathbb{Q}_l(\mu_l)^\times \rightarrow (\mathcal{O}_{\mathfrak{l}}/(1-\zeta_l))^\times$ given by $x \mapsto x^{1-\sigma_l}$ (where σ_l is a generator of $\text{Gal}(\mathbb{Q}_l(\mu_l)/\mathbb{Q}_l)$). Since the extension is totally, tamely ramified, $G_0 = I_l = \text{Gal}(\mathbb{Q}_l(\mu_l)/\mathbb{Q}_l)$ and $G_1 = 1$ where G_i denote ramification groups. This implies that σ_l has trivial action on $\mathcal{O}_{\mathfrak{l}}/(1-\zeta_l)$ and nontrivial action on $\mathcal{O}_{\mathfrak{l}}/(1-\zeta_l)^2$. In particular, σ_l has transitive action on the cosets $k(1-\zeta_l) + (1-\zeta_l)^2\mathcal{O}_{\mathfrak{l}}$ so the map $x \mapsto x^{1-\sigma_l}$ is surjective locally. From this, we get that globally the map surjects onto $(\mathcal{O}_K/l\mathcal{O}_K)^\times$. Since $l \equiv 1 \pmod{M}$, the kernel of the left map is contained in the kernel of the right map, so the right map factors through the left map. $\blacksquare$

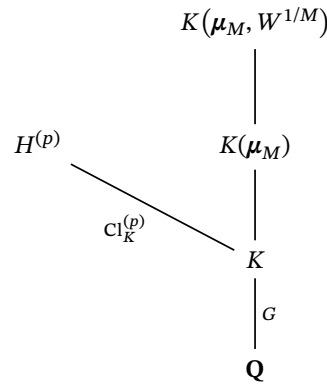
Theorem 2: (Rubin)

Let $p > 2$ be a rational prime not dividing $[K : \mathbf{Q}]$ and $\text{Cl}_K^{(p)}$ the p -part of the ideal class group of K . Suppose one is given $\mathfrak{c} \in \text{Cl}_K^{(p)}$, $M = p^k$ for some fixed k ; for $G = \text{Gal}(K/\mathbf{Q})$, a finite G -submodule W of $K^\times / (K^\times)^M$, and a G -equivariant map $\psi : W \rightarrow (\mathbf{Z}/M\mathbf{Z})[G]$.

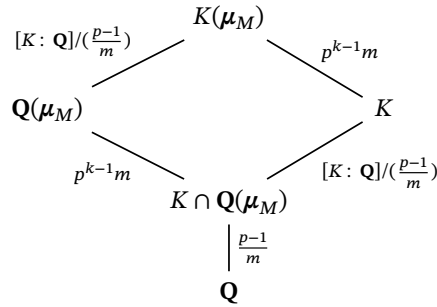
Then there are infinitely many primes λ of K such that

- i. $\lambda \in \mathfrak{c}$
- ii. For $l = \lambda \cap \mathbf{Z}$, $l \in I_M^1$
- iii. $[w]_l = 0$ for all $w \in W$, and there is a $u \in (\mathbf{Z}/M\mathbf{Z})^\times$ such that $\varphi_l(w) = u\psi(w)\lambda$ where φ_l is the map from [lemma 3](#).

Proof. Let $H^{(p)}$ denote the p -Hilbert class field of K i.e. the maximal abelian unramified p -extension of K ; then $\text{Gal}(H^{(p)}/K) \cong \text{Cl}_K^{(p)}$ and we have the diagram below.



We follow [Rub90] and first show that $K(\mu_M) \cap H^{(p)} = K$. We note that $K(\mu_M)/K$ can be ramified only at p and ∞ , so it suffices to show that primes over p must ramify in any p -subextension of $K(\mu_M)/K$. We have the following lattice of fields:



Since $Q(\mu_M)/\mathbf{Q}$ is totally ramified at p , the extension $K(\mu_M)/\mathbf{Q}$ must also be ramified at p , with ramification index divisible by p^{k-1} . Since $p \nmid [K : \mathbf{Q}]$, the extension $K/\mathbf{Q}$ has no p -power index ramification at p ; it thus follows that the extension $K(\mu_M)/K$ is ramified at p with ramification index divisible by p^{k-1} , so any p -subextension must be ramified at p .

We note that W can be identified as a subgroup of $K(\mu_M)^\times / (K(\mu_M)^\times)^M$; indeed, the Kummer exact sequence

$$1 \longrightarrow \mu_n \longrightarrow \bar{L}^\times \xrightarrow{\cdot n} \bar{L}^\times \longrightarrow 1$$

yields that $L^\times / (L^\times)^n = H^1(L, \mu_n)$ for any field L and any $n \in \mathbf{N}$. The inflation-restriction exact sequence yields

$$0 \longrightarrow H^1(K(\mu_M)/K, \mu_M^{G_{K(\mu_M)}}) \xrightarrow{\text{inf}} H^1(K, \mu_M) \xrightarrow{\text{res}} H^1(K(\mu_M), \mu_M)$$

since $\mu_M^{G_{K(\mu_M)}} = 1$, the restriction map is injective. We thus have the inclusions

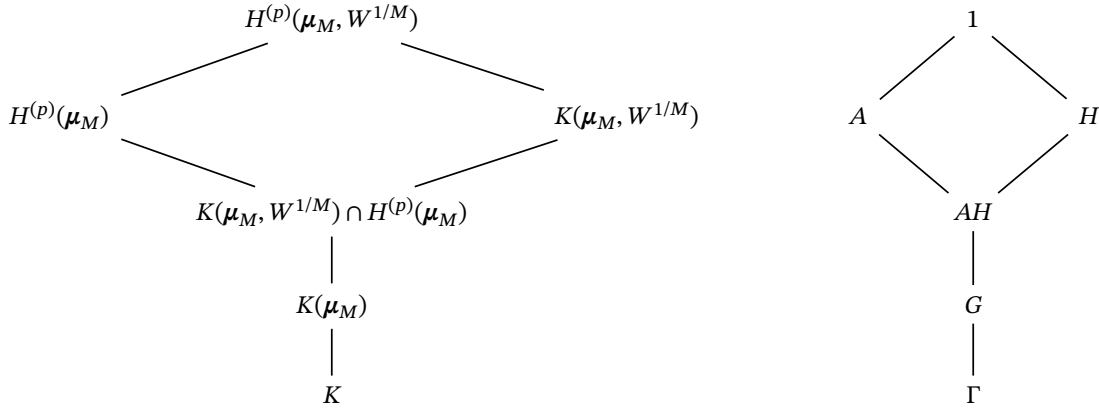
$$W \hookrightarrow K^\times / (K^\times)^M \xrightarrow{\text{res}} K(\mu_M)^\times / (K(\mu_M)^\times)^M$$

as desired.

We claim now that $K(\mu_M, W^{1/M}) \cap H^{(p)} = K$; we first note that since $W \subset K(\mu_M)^\times / (K(\mu_M)^\times)^M$, the Kummer pairing

$$\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M)) \times W \rightarrow \mu_M$$

yields the isomorphism $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M)) = \text{Hom}(W, \mu_M)$. Since $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$ is a normal subgroup of $\text{Gal}(K(\mu_M, W^{1/M})/K)$, there is an induced action of $\text{Gal}(K(\mu_M)/K)$ on $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$ via conjugation. In particular, let σ denote complex conjugation in $\text{Gal}(K(\mu_M)/K) \subset \text{Gal}(K(\mu_M)/\mathbb{Q})$; since the Kummer pairing is $\text{Gal}(K(\mu_M)/\mathbb{Q})$ -equivariant, the action of σ on $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$ is given by the action on $\text{Hom}(W, \mu_M)$. Since $W \subset K^\times / (K^\times)^M$, σ acts trivially on W ; as σ acts as -1 on μ_M , it follows that σ acts as -1 on $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$. Since $H^{(p)}$ is an abelian extension of K and $H^{(p)} \cap K(\mu_M) = K$ from before, we know that $\text{Gal}(H^{(p)}(\mu_M)/K)$ is abelian. In particular we must have that the action of σ (this time viewed in the quotient of $\text{Gal}(H^{(p)}(\mu_M)/K)$ on $\text{Gal}(H^{(p)}(\mu_M)/K(\mu_M))$ must be trivial. We have the following lattice of fields with the corresponding Galois groups:



In particular, the conjugation action of Γ/G on G/A and G/H must agree on G/AH so σ acts by 1 and -1 on $\text{Gal}(K(\mu_M, W^{1/M}) \cap H^{(p)}(\mu_M)/K(\mu_M))$. This group must then be 2 -torsion, but since it is a p -group it must be trivial; thus $K(\mu_M, W^{1/M}) \cap H^{(p)}(\mu_M) = K(\mu_M)$. Intersecting both sides with $H^{(p)}$ yields

$$K(\mu_M, W^{1/M}) \cap H^{(p)} = K(\mu_M) \cap H^{(p)} = K.$$

We now fix a primitive M^{th} root of unity ζ_M and define $f: (\mathbb{Z}/M\mathbb{Z})[G] \rightarrow \mu_M$ by $f(1) = \zeta_M$ and $f(g) = 1$ for non-identity $g \in G$. Let $\gamma \in \text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$ correspond to $f \circ \psi \in \text{Hom}(W, \mu_M)$ under the Kummer isomorphism. Then by the definition of the Kummer pairing we have $f \circ \psi(w) = (w^{1/M})^\gamma / w^{1/M}$ for all $w \in W$. Since $K(\mu_M, W^{1/M}) \cap H^{(p)} = K$, we can choose $\tau \in \text{Gal}(H^{(p)}(\mu_M, W^{1/M})/K)$ such that τ restricts to γ on $K(\mu_M, W^{1/M})$ and to c on $H^{(p)}$. By the Chebotarev Density Theorem, there exist infinitely primes λ of K whose Frobenius in $\text{Gal}(H^{(p)}(\mu_M, W^{1/M})/\mathbb{Q})$ is the conjugacy class of τ . We claim that these λ satisfy the theorem.

For each λ , compatibility of Frobenius elements implies that the Frobenius of λ in $\text{Gal}(H^{(p)}/K)$ is c meaning i. is satisfied. Similarly, since γ is trivial on $K(\mu_M)$ the Frobenius of λ is the identity. Thus for l the rational prime below λ , l splits completely in $K(\mu_M)$ and thus in $\mathbb{Q}(\mu_M)$ meaning ii. is satisfied. For the first assertion of iii., note that each $w \in W$ is an M^{th} power in $K(\mu_M, W^{1/M})$. Since λ by construction is unramified in this extension, it follows that $[w]_l = 0$.

For the second assertion of iii., note that $v_\lambda(\varphi_l(w)) = 0$ if and only if w is an M^{th} power modulo λ . Further, we have the following chain of equivalences:

$$v_\lambda(\lambda^{\psi(w)}) = 0 \iff f \circ \psi(w) = 1 \iff \frac{(w^{1/M})^\gamma}{w^{1/M}} = 1 \iff w \text{ is an } M^{\text{th}} \text{ power modulo } \lambda.$$

Thus there exists $u \in (\mathbf{Z}/M\mathbf{Z})^\times$ such that $v_\lambda(\varphi_l(w)) = uv_\lambda(\lambda^{\psi(w)})$ for all $w \in W$. It follows that the map $w \mapsto \varphi_l(w) - u\psi(w)\lambda$ is a $\text{Gal}(K/\mathbf{Q})$ -equivariant homomorphism $W \rightarrow \bigoplus_{\mathfrak{l}|\mathfrak{l}, \mathfrak{l} \neq \lambda} (\mathbf{Z}/M\mathbf{Z})\mathfrak{l}$. The codomain has no nontrivial Galois-stable submodules, so this must be the zero map. Thus $\varphi_l(w) = u\psi(w)\lambda$ as claimed. ■

Remark:

It is possible to remove the assumption that $p \nmid [K : \mathbf{Q}]$ in the case that $K = \mathbf{Q}(\mu_n)^+$ is the maximal real subfield of a cyclotomic field –this is the case treated in theorem 3.1 of [Rub90].

4 The Main Result

Let χ be a p -adic valued Dirichlet character of $\text{Gal}(K/\mathbf{Q})$; we write $\mathbf{Z}_p[\chi]$ (resp. $\mathbf{Q}_p(\chi)$) to mean $\mathbf{Z}_p[\chi(\text{Gal}(K/\mathbf{Q}))]$. We define the idempotent

$$e(\chi) := \frac{1}{[K:\mathbf{Q}]} \sum_{\sigma \in \text{Gal}(K/\mathbf{Q})} \text{Tr}(\chi(\sigma)^{-1})\sigma$$

where Tr denote the field-theoretic trace from $\mathbf{Q}_p(\chi)$ to $\mathbf{Q}_p$. We then have $e(\chi) \in \mathbf{Z}_p[\chi]$, so for a $\mathbf{Z}[\text{Gal}(K/\mathbf{Q})]$ -module of finite type A , we define

$$A(\chi) := e(\chi)(A \otimes_{\mathbf{Z}} \mathbf{Z}_p).$$

Then $A(\chi)$ is a $\mathbf{Z}_p[\text{Gal}(K/\mathbf{Q})]$ -module and we have

$$A \otimes_{\mathbf{Z}} \mathbf{Z}_p \cong \bigoplus_{\chi} A(\chi),$$

where the direct sum runs over representatives of all $G_{\mathbf{Q}_p}$ -orbits of characters of $\text{Gal}(K/\mathbf{Q})$.

Fix an odd rational prime p and let M be a power of p divisible by the p -class number of K . From the natural maps $E_K/E_K^M \rightarrow E_K/E_K^M C_K \cong (E_K/C_K) \otimes_{\mathbf{Z}} \mathbf{Z}_p$ and the χ -eigenspace decomposition, it follows that there exists $\varepsilon \in E_K$ such that εC_K generates $(E_K/C_K)(\chi)$ [All17, Lemma 2.2],[All13, Theorem 1.11]. We define I_{ε}^n to be the subset of I_M^n with the following properties:

1. Every prime divisor l of $r \in I_{\varepsilon}^n$ splits completely over the splitting field of $x^p - \varepsilon$;
2. There is a prime $\mathfrak{l}$ of K over l with ideal class in the eigenspace $\text{Cl}_K(\chi)$.

As in [Seo08], one may define $\mathcal{E}_{\varepsilon}^n$ and S_{ε}^n in a similar manner to $\mathcal{E}^n$ and S^n .

For a prime $l \in I_M^1$ (so $l \equiv 1 \pmod{M}$), we write $l-1 = p^{n_l} m_l$ where $(m_l, p) = 1$.

Lemma 4

For any prime $\mathfrak{l}$ of K over l , ε^{m_l} generates the p -Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^{\times}$ if and only if l does not split completely over the splitting field of $x^p - \varepsilon$.

Proof. Suppose l splits completely over the splitting field of $x^p - \varepsilon$. From this we see immediately that any $\mathfrak{l} \mid l$ splits completely in the extension $K(\sqrt[p]{\varepsilon})/K$. As $\varepsilon \in \mathcal{O}_K^{\times}$, $\mathfrak{l} \nmid \text{disc } \mathcal{O}_K[\sqrt[p]{\varepsilon}]$. Thus $\mathfrak{l}$ splits completely in $\mathcal{O}_K[\sqrt[p]{\varepsilon}]$, so $x^p - \varepsilon$ splits completely over $\mathcal{O}_K/\mathfrak{l}$. This implies that ε is a p^{th} power mod $\mathfrak{l}$, as is ε^{m_l} . Then ε^{m_l} cannot generate the p -Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^{\times}$.

Conversely, suppose ε^{m_l} does not generate the p -Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^{\times}$ for any $\mathfrak{l} \mid l$; we must have that ε^{m_l} is a p^{th} power mod $\mathfrak{l}$. As $(p, m_l) = 1$, ε must also be a p^{th} power mod $\mathfrak{l}$. Since $l \equiv 1 \pmod{p}$ and $\mathcal{O}_K/\mathfrak{l} \cong \mathbf{Z}/l$, we obtain that $x^p - \varepsilon$ splits completely over $\mathcal{O}_K/\mathfrak{l}$. This implies that $\mathfrak{l}$ splits completely in $\mathcal{O}_K[\sqrt[p]{\varepsilon}]$; since $\varepsilon \in E_K$ we also obtain that each $\mathfrak{l}$ (and thus l) splits completely in $K(\sqrt[p]{\varepsilon})$. l then must also split completely over the Galois closure of $K(\sqrt[p]{\varepsilon})$; that is, l splits completely over the splitting field of $x^p - \varepsilon$. ■

Fix $\mathfrak{c} \in \text{Cl}_K(\chi)$; define $I_p^1(\mathfrak{c})$ as the primes l in I_p^1 such that some prime $\mathfrak{l}$ in K over l is contained in the ideal class $\mathfrak{c}$. Let ε^{p^s} denote the map in $\mathcal{E}^1$ which maps $1 \mapsto \varepsilon^{p^s}$ and $l \mapsto 1$ for any prime l . Set

$$W := \langle \varepsilon^{p^s}(\chi) \rangle \subset K^{\times}/(K^{\times})^M.$$

Theorem 3

Let l be a rational prime which splits completely in $K(\mu_M)$ and is completely inert in the extension $K(\mu_M, W^{1/M})/K(\mu_M)$. For any prime $\mathfrak{l}$ of K over l , ε^{m_l} generates the p -Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^{\times}$. Moreover, if $p \nmid [K:\mathbf{Q}]$ then $I_p^1(\mathfrak{c}) \setminus I_{\varepsilon}^1$ is infinite for any $\mathfrak{c} \in \text{Cl}_K(\chi)$.

Proof. Suppose for contradiction that l satisfies the conditions of the theorem and ε^{m_l} does not generate the p -Sylow subgroup of $(\mathcal{O}_K/\mathfrak{l})^\times$. By [lemma 4](#), l splits completely over $K(\sqrt[p]{\varepsilon}, \mu_p)$, the splitting field of $x^p - \varepsilon$. From the definitions of W and M , $K(\mu_p, \sqrt[p]{\varepsilon}) \subset K(\mu_M, W^{1/M})$. We claim that in fact $K(\mu_p, \sqrt[p]{\varepsilon}) \subset K(\mu_M)$. We have the following tower of fields and the corresponding lattice of Galois groups:

$$\begin{array}{ccc}
 K(\mu_M, W^{1/M}) = T_l & & 1 = I_l \\
 \downarrow & & \downarrow \\
 K(\mu_M) = Z_l & & \text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M)) = G_l \\
 \downarrow & & \downarrow \\
 K & & \text{Gal}(K(\mu_M, W^{1/M})/K) \\
 \downarrow & & \downarrow \\
 \mathbb{Q} & & \text{Gal}(K(\mu_M, W^{1/M})/\mathbb{Q})
 \end{array}$$

The Frobenius element of l in $K(\mu_M, W^{1/M})$ is a generator of $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$. As l splits completely in $K(\mu_p, \sqrt[p]{\varepsilon})$, the Frobenius of l must restrict to the identity element on this field. This implies that $K(\mu_p, \sqrt[p]{\varepsilon})$ is fixed by $\text{Gal}(K(\mu_M, W^{1/M})/K(\mu_M))$, so $K(\mu_p, \sqrt[p]{\varepsilon}) \subset K(\mu_M)$ as claimed. This implies that $K(\sqrt[p]{\varepsilon})$ is a Galois extension of K , meaning either $\mu_p \subset K^\times$ or $\sqrt[p]{\varepsilon} \in K$. As K is real, the former cannot be the case, and as ε generates $(E_K/S_\varepsilon^1)(\chi)$, the latter is impossible as well.

For the second claim, note that $l \in I_p^1$ splits completely in K and is 1 mod M , so l splits completely in $K(\mu_M)$. Further, since W is a cyclic submodule of $K^\times/(K^\times)^M$, $K(W^{1/M}) = K(\varepsilon^{p^s/M})$. Kummer theory then tells us that $K(\mu_M, W^{1/M})/K(\mu_M)$ is a cyclic extension of order dividing M . Thus any prime having a generator of $\text{Gal}(K(\mu_M, W^{1/M}), K(\mu_M))$ as its Frobenius element satisfies the conditions of the theorem, and as a result will not split completely in $K(\mu_p, \sqrt[p]{\varepsilon})$. From the proof of [theorem 2](#), we know that we can choose $\tau \in \text{Gal}(H^{(p)}(\mu_M, W^{1/M})/K)$ which restricts to ϵ in $\text{Gal}(H^{(p)}/K)$ and a chosen generator of $\text{Gal}(K(\mu_M, W^{1/M}), K(\mu_M))$ on $K(\mu_M, W^{1/M})$. Any prime of K having the conjugacy class of τ as its Frobenius will then be in $I_p^1 \setminus I_\varepsilon^1$, and there are infinitely many such primes by the Chebotarev Density Theorem. ■

5 Next Steps

Having developed some of the basic machinery of Euler systems and the theory of special units, there are numerous open avenues for mathematical development. Primarily is the following conjecture of Seo (stated in [Seo08]):

Conjecture 1: (Seo)

Let U be the set of all primes p such that $|\text{Cl}_K^{(p)}| \neq |(E_K/C_K)^{(p)}|$. Define $\mathbf{Z}'$ to be the smallest subring of $\mathbf{Q}$ containing $\mathbf{Z}$, $\frac{1}{2}$, and p^{-1} for all $p \in U$. Then

$$\text{Cl}_K \cong \text{gr}(S_K) \text{ as } \mathbf{Z}'[\text{Gal}(K/\mathbf{Q})]\text{-modules,}$$

where

$$\text{gr}(S_K) := \bigoplus_{n \geq 0} S_K^n / S_K^{n+1}.$$

We note that Seo's conjecture in particular implies the following:

Conjecture 2

For an abelian group G , let $\exp(G)$ denote the exponent of G i.e. the smallest $n \in \mathbf{Z}_{>0}$ which annihilates G . Then

$$\exp(\text{Cl}_K) = [E_K : S_K^1].$$

Proofs of these conjectures would be major results in the study of the ideal class group, and may have significant influences on other open questions such as Gauss' real class number one problem, which asks whether there are infinitely many real quadratic fields with class number one⁷.

⁷It is not even known whether there are infinitely many number fields with this property.

A Algebraic Number Theory

We summarise here the basic notions of algebraic number theory. For more details see [Mar18] or [Neu99].

For any number field K (finite extension of $\mathbf{Q}$), we define the ring of integers $\mathcal{O}_K$ to be the integral closure of $\mathbf{Z}$ in K i.e. those $\alpha \in K$ which are roots to monic polynomials in $\mathbf{Z}[x]$. The ideals of $\mathcal{O}_K$ factor uniquely (up to order of factors) into products of prime ideals, so $\mathcal{O}_K$ is always a Dedekind domain.

For any number field K we define the ideal class group Cl_K to be the ideals of $\mathcal{O}_K$, up to the equivalence

$$\mathfrak{a} \equiv \mathfrak{b} \iff \alpha \mathfrak{a} = \beta \mathfrak{b} \text{ for some } \alpha, \beta \in \mathcal{O}_K.$$

This forms an abelian group under the operation $[\mathfrak{a}][\mathfrak{b}] = [\mathfrak{a}\mathfrak{b}]$. We can make an alternative definition of Cl_K as the group of fractional ideals J_K modulo the principal fractional ideals P_K . This construction is encoded in the following exact sequence:

$$1 \longrightarrow \mathcal{O}_K^\times \longrightarrow K^\times \longrightarrow J_K \longrightarrow \text{Cl}_K \longrightarrow 0.$$

The two most essential properties of the class group are given in the following:

Theorem: Triviality of the class group

Let K be a number field. Then $\text{Cl}_K = 1$ if and only if $\mathcal{O}_K$ is a unique factorization domain.

Theorem: Finiteness of the class group

Let K be a number field. Then the ideal class group Cl_K is finite.

We also have the following essential theorem on the structure of $\mathcal{O}_K^\times$:

Theorem: Dirichlet's Unit Theorem

Let K be a number field, and let r_1 denote the number of real embeddings of K and r_2 the number of *pairs* of complex embeddings of K (so $r_1 + 2r_2 = [K : \mathbf{Q}]$). Then we have the following decomposition of the global units of K :

$$E_K = \mathcal{O}_K^\times \cong W_K \times \mathbf{Z}^{r_1 + r_2 - 1}$$

where W_K is the roots of unity in K and is finite.

Let L/K be an extension of number fields. For a prime $\mathfrak{p}$ of K (a prime ideal of $\mathcal{O}_K$), we can consider its factorization into primes of L :

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}.$$

We say $\mathfrak{p}$ is *ramified* in L if any e_i is larger than 1, and that $\mathfrak{p}$ is unramified otherwise. For each $\mathfrak{P}_i$ over $\mathfrak{p}$ we write $f_i = f(\mathfrak{P}_i|\mathfrak{p})$ to be the degree of the residue field extension $[\mathcal{O}_L/\mathfrak{P}_i : \mathcal{O}_K/\mathfrak{p}]$. We have the following formula:

$$[L : K] = \sum_{i=1}^r e_i f_i.$$

If L/K is a Galois extension, then $\text{Gal}(L/K)$ acts transitively on the primes $\mathfrak{P}_i$ over $\mathfrak{p}$. This implies that $f_1 = \cdots = f_r$ and $e_1 = \cdots = e_r$, and we can write e and f to get the simple formula $[L : K] = ref$. For each prime $\mathfrak{P}$ over $\mathfrak{p}$ we can define the *decomposition group* $G_{\mathfrak{P}} \subset \text{Gal}(L/K)$ as the stabilizer of $\mathfrak{P}$ under the Galois action. Let λ denote the residue field $\mathcal{O}_L/\mathfrak{P}$ and $\kappa = \mathcal{O}_K/\mathfrak{p}$. There is a homomorphism $G_{\mathfrak{P}} \rightarrow \text{Gal}(\lambda/\kappa)$ induced by reduction mod $\mathfrak{P}$, and we call the kernel $I_{\mathfrak{P}}$ of this homomorphism the *inertia group*. If $\mathfrak{P}_1, \mathfrak{P}_2$ are primes over $\mathfrak{p}$ then the groups $G_{\mathfrak{P}_1}$ and $G_{\mathfrak{P}_2}$ (resp. $I_{\mathfrak{P}_1}$ and $I_{\mathfrak{P}_2}$) are conjugates. Thus when L/K is an abelian extension these groups are the same, and we can write $G_{\mathfrak{p}}$ and $I_{\mathfrak{p}}$ instead.

For any subgroup H of $\text{Gal}(L/K)$ we let L^H denote the fixed field of H . We define the decomposition and inertia fields, $Z_{\mathfrak{p}} = L^{G_{\mathfrak{p}}}$ and $T_{\mathfrak{p}}$, respectively. We have the following lattice of fields with the associated Galois groups:

$$\begin{array}{ccc}
 L & & 1 \\
 e \downarrow & & \downarrow e \\
 T_{\mathfrak{p}} & & I_{\mathfrak{p}} \\
 f \downarrow & & \downarrow f \\
 Z_{\mathfrak{p}} & & G_{\mathfrak{p}} \\
 r \downarrow & & \downarrow r \\
 K & & \text{Gal}(L/K)
 \end{array}$$

In particular, When $\mathfrak{p}$ is unramified in the extension L/K , we get that $I_{\mathfrak{p}} = 1$ so the map $G_{\mathfrak{p}} \rightarrow \text{Gal}(\lambda/\kappa)$ is an isomorphism. The Frobenius automorphism of λ acts as exponentiation by $|\kappa|$ and is a generator of $\text{Gal}(\lambda/\kappa)$. We denote $\text{Fr}_{\mathfrak{p}}$ to be the element of $G_{\mathfrak{p}}$ corresponding to the Frobenius automorphism in $\text{Gal}(\lambda/\kappa)$ and call this the *Frobenius of $\mathfrak{p}$* . As before, when L/K is abelian we may consider the Frobenius of $\mathfrak{p}$, $\text{Fr}_{\mathfrak{p}}$, instead.

The following theorem is an “explicit class field theory” for $\mathbf{Q}$, and is used quite frequently in the study of abelian number fields.

Theorem: Kronecker-Weber

Let K be an abelian extension of $\mathbf{Q}$. Then $K \subset \mathbf{Q}(\zeta_n)$ for some n .

The following theorem is also used frequently, and may be thought of as a generalization of Dirichlet’s theorem on primes in arithmetic progressions:

Theorem: Chebotarev Density

Let L/K be a Galois extension of number fields and let C be a subset of $\text{Gal}(L/K)$ which is stable under conjugation. Define S to be the set of primes $\mathfrak{p}$ of K with $\text{Fr}_{\mathfrak{p}} \in C$ for some $\mathfrak{P} \mid \mathfrak{p}$. Then the Dirichlet density of S is $|C|/[L : K]$. In particular, there are infinitely many primes of K in S .

References

- [All13] Timothy J. All. *On the Galois module structure of the units and ray classes of a real abelian number field*. PhD thesis, The Ohio State University, 2013.
- [All17] Timothy J. All. Gauss sums, Stickelberger’s theorem and the Gras conjecture for ray class groups. *Acta Arithmetica*, 173(3):273 – 299, 2017.
- [BCDT01] Christopher Breuil, Brian Conrad, Fred Diamond, and Richard Taylor. On the modularity of elliptic curves over $\mathbf{Q}$: wild 3-adic exercises. *Journal of the American Mathematical Society*, 14:843 – 939, 2001.
- [Fre86] Gerhard Frey. Links between stable elliptic curves and certain diophantine equations. *Annales Universitatis Saraviensis*, 1:1 – 40, 1986.
- [Iwa59] Kenkichi Iwasawa. On Γ -extensions of algebraic number fields. *Bulletin of the American Mathematical Society*, 65:183 – 226, 1959.
- [Ked] Kiran S. Kedlaya. Notes on class field theory. <https://kskedlaya.org/cft>.
- [Kol90] Victor Kolyvagin. Euler systems. *The Grothendieck Festschrift*, 2:435 – 483, 1990.
- [Mar18] Daniel A. Marcus. *Number Fields*. Springer-Verlag, 2018.
- [MW84] Barry Mazur and Andrew Wiles. Class fields of abelian extensions of $\mathbf{Q}$. *Inventiones mathematicae*, 76:179 – 330, 1984.
- [Neu99] Jürgen Neukirch. *Algebraic Number Theory*. Springer-Verlag, 1999.
- [NSW13] Jürgen Neukirch, Alexander Schmidt, and Kay Wingberg. *Cohomology of Number Fields*. Springer-Verlag, 2013.
- [Rib90] Kenneth A. Ribet. On modular representations of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ arising from modular forms. *Inventiones mathematicae*, 100:431 – 476, 1990.
- [Rub87] Karl Rubin. Global units and ideal class groups. *Inventiones mathematicae*, 89:511–526, 1987.
- [Rub90] Karl Rubin. *The main conjecture, Appendix to the second edition of S. Lang: Cyclotomic Fields*. Springer-Verlag, 1990.
- [Seo08] Soogil Seo. Truncated Euler systems. *Journal für die reine und angewandte Mathematik*, 614:53 – 71, 2008.
- [Sin80] Warren M. Sinnott. On the Stickelberger Ideal and the Circular Units of an Abelian Field. *Inventiones mathematicae*, 62(2):181–234, 1980.
- [Tha88] Francisco Thaine. On the ideal class groups of real abelian number fields. *Annals of Mathematics*, 128:1 – 18, 1988.
- [TW95] Richard Taylor and Andrew Wiles. Ring-theoretic properties of certain Hecke algebras. *Annals of Mathematics*, 141:553 – 572, 1995.
- [Was82] Lawrence C. Washington. *Introduction to Cyclotomic Fields*. Springer-Verlag, 1982.
- [Wil95] Andrew John Wiles. Modular elliptic curves and Fermat’s Last Theorem. *Annals of Mathematics*, 141:443 – 551, 1995.